

JAR — GROWTH INTERN ASSIGNMENT

Sales Analysis · App Exploration · Product Strategy

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Analysis performed in Python (pandas) — full script attached as app.py

Question 1 — Sales Analysis

Part 1: Sales & Profitability Analysis by Category

The List of Orders and Order Details datasets were merged on Order ID, producing a clean 1-to-many join with no unmatched records (500 orders → 1,500 line items). Category-level metrics were then computed on the merged table:

Category	Total Sales (₹)	Total Profit (₹)	Avg. Profit / Order (₹)	Profit Margin (%)
Electronics	1,65,267	10,494	34.07	6.35%
Clothing	1,39,054	11,163	11.76	8.03%
Furniture	1,27,181	2,298	9.46	1.81%

Table 1 — Category-wise sales and profitability (source: merged Order data, 1,500 line items)

Category-wise Total Sales vs Profit Margin

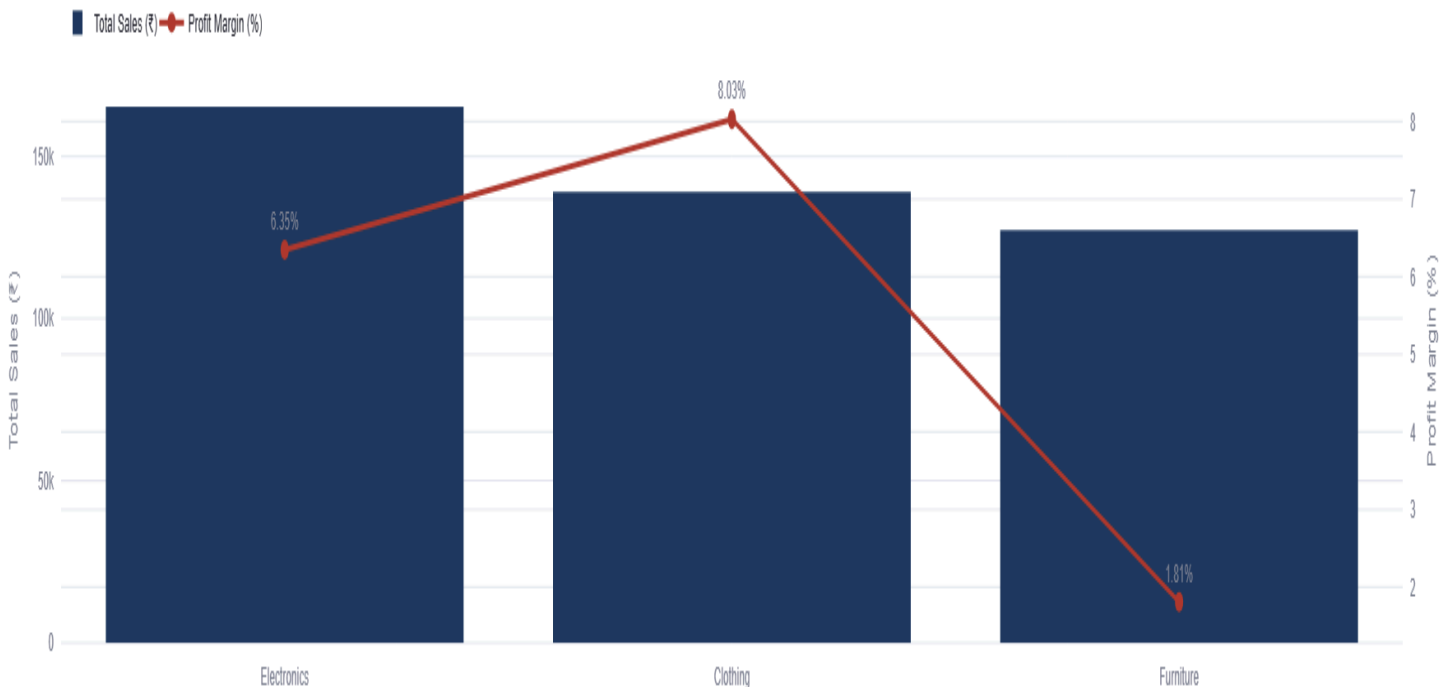


Figure 1 — Total sales (bars) vs profit margin (line) by category

Top performer: Clothing (best margin) & Electronics (best absolute profit)

- **Clothing** delivers the healthiest margin at 8.03%, despite generating the lowest per-order profit of the three (₹11.76/order). It sells in high volume (949 line items, nearly 3× Furniture's) and every sub-category except none is profitable — even its weakest performer (Saree, 0.66% margin) stays in positive territory. This is a classic low-ticket, high-frequency, low-risk category.

- **Electronics** generates the highest total profit (₹10,494) and the best profit-per-order (₹34.07) — nearly 3.6× Clothing's — because it carries a higher average selling price. However, its margin (6.35%) sits in the middle because one sub-category, Electronic Games, is loss-making.

Underperformer: Furniture (weakest on every profitability metric)

- **Furniture** is the clear underperformer: total profit of only ₹2,298 on ₹1,27,181 of sales — a 1.81% margin, roughly 4–5× thinner than the other two categories.

Category	Loss-making Sub-Category	Sales (₹)	Profit (₹)	Margin
Furniture	Tables	22,614	-4,011	-17.74%
Electronics	Electronic Games	39,168	-1,236	-3.16%

Table 2 — The two sub-categories operating at a net loss

Tables is the single biggest drag on the entire dataset: at -17.74% margin it wipes out most of the gains from Furniture's next-best performer (Bookcases, 8.6% margin, ₹4,888 profit). This one sub-category alone explains why Furniture underperforms.

Likely reasons for the performance gap

- Discounting pressure on big-ticket, low-frequency items: Tables and large Furniture pieces are typically sold with negotiated or promotional discounts to close the sale, which erodes margin far more than it does on small accessory items.
- Higher logistics cost-to-value ratio: bulky, fragile items (tables, bookcases, chairs) carry disproportionately high shipping, packaging, and damage/return costs relative to their sale price, unlike compact Clothing or Electronic accessories.
- Category-specific demand elasticity: Electronic Games appear to be a promotional/loss-leader line (bundled offers, MRP-matching against e-commerce competitors) that drives footfall/cross-sell rather than standalone profit.
- Portfolio mix within Clothing skews toward small, high-margin accessories (Stole, Hanky, T-Shirt at 13–20% margin) that offset the low-margin Saree line, giving the category its resilience.

Recommendation

- Re-price or bundle Tables (e.g., attach assembly/delivery as a paid add-on rather than absorbing it in the product price) and audit discount slabs specifically for this sub-category.
- Investigate Electronic Games' pricing — if it is intentionally run as a loss-leader, measure whether it is actually lifting attach-rate sales of Phones/Accessories/Printers; if not, reprice it.
- Double down on high-margin Clothing accessories (T-Shirt, Stole, Hankerchief) and Electronics Accessories/Printers (16.4% and 10.2% margin) as the categories to prioritise for marketing spend.

Part 2: Target Achievement Analysis — Furniture

The Sales Target dataset provides 12 monthly Furniture targets covering one full sales cycle (April → March). Month-over-month percentage change was calculated on the target values:

Month	Furniture Target (₹)	MoM % Change
Apr	10,400	—
May	10,500	+0.96%
Jun	10,600	+0.95%
Jul	10,800	+1.89%
Aug	10,900	+0.93%
Sep	11,000	+0.92%
Oct	11,100	+0.91%
Nov	11,300	+1.80%
Dec	11,400	+0.88%
Jan	11,500	+0.88%
Feb	11,600	+0.87%
Mar	11,800	+1.72%

Table 3 — Furniture monthly target and month-over-month % change

Part 2: Target Achievement Analysis — Furniture

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Furniture Monthly Target Trend (Apr → Mar)

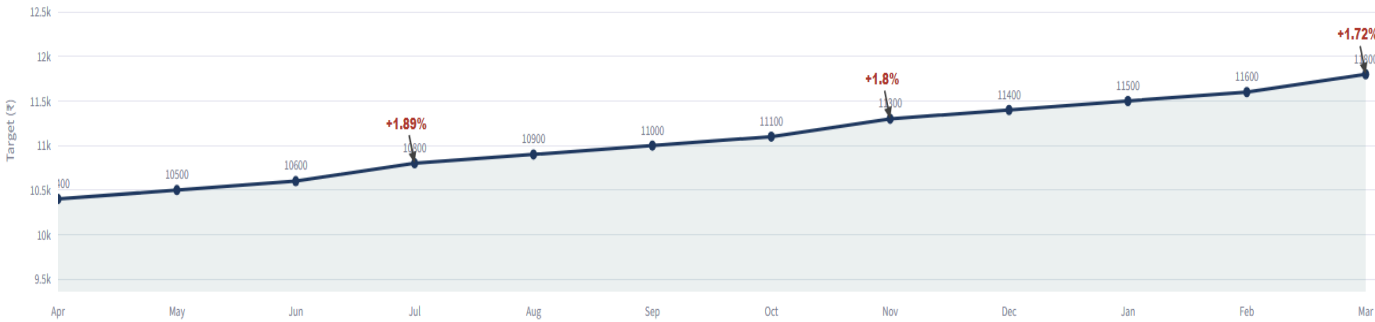


Figure 2 — Furniture target trend across the sales cycle

Trend analysis

- **Overall pattern is smooth, not volatile:** targets rise steadily by roughly 0.9–1% almost every month, growing from ₹10,400 in April to ₹11,800 in March — a cumulative +13.5% over the full cycle, with no single month showing an erratic spike or cut.
- **A recurring quarterly step-up:** the three largest increases — July (+1.89%), November (+1.80%), and March (+1.72%) — each fall at the end of a quarter (Q1, Q3, Q4 in an Apr–Mar calendar). This suggests targets are re-baselined quarterly rather than purely linearly, likely to account for seasonal demand (festive/wedding-season furniture buying around Oct–Nov, and financial-year-end sales pushes in March).
- **No genuine 'fluctuation' risk:** because every month is a positive increment, the risk here isn't sudden target swings — it's whether steadily rising targets are matched by steadily rising actual sales, especially since Furniture already has the thinnest margin (1.81%) of the three categories from Part 1.

Strategies to align targets with actual performance

- Tie the quarterly step-ups (Jul/Nov/Mar) explicitly to seasonal campaigns — e.g., pre-festive inventory pushes and bundled discounts timed to the same month the target jumps, so the sales team has a corresponding lever, not just a bigger number.
- Set the Furniture target with a margin-adjusted view, not just revenue — since the category's profit margin is thin (1.81%) and Tables specifically runs at a loss, a pure sales-target chase can hit the ₹ number while profitability keeps falling.
- Use a rolling 3-month actual-vs-target variance review (rather than a single annual check) so the ~1% monthly creep doesn't quietly compound into a large year-end miss that's only caught in March.
- Set sub-category-level sub-targets within Furniture (e.g., protect/grow Bookcases at 8.6% margin, cap or restructure Tables) so hitting the category target doesn't come disproportionately from the loss-making line.

Part 3: Regional Performance Insights

From the List of Orders dataset (merged with Order Details for sales/profit figures), the states with the highest order counts were identified:

State	Order Count	Total Sales (₹)	Total Profit (₹)	Avg. Profit/Order (₹)	Margin
Madhya Pradesh	101	1,05,140	5,551	54.96	5.28%
Maharashtra	90	95,348	6,176	68.62	6.48%
Rajasthan	32	21,149	1,257	39.28	5.94%
Gujarat	27	21,058	465	17.22	2.21%
Punjab	25	16,786	−609	−24.36	−3.63%

Table 4 — Top 5 states by order count with sales & profit performance

Part 3: Regional Performance Insights

Top 5 States by Order Count

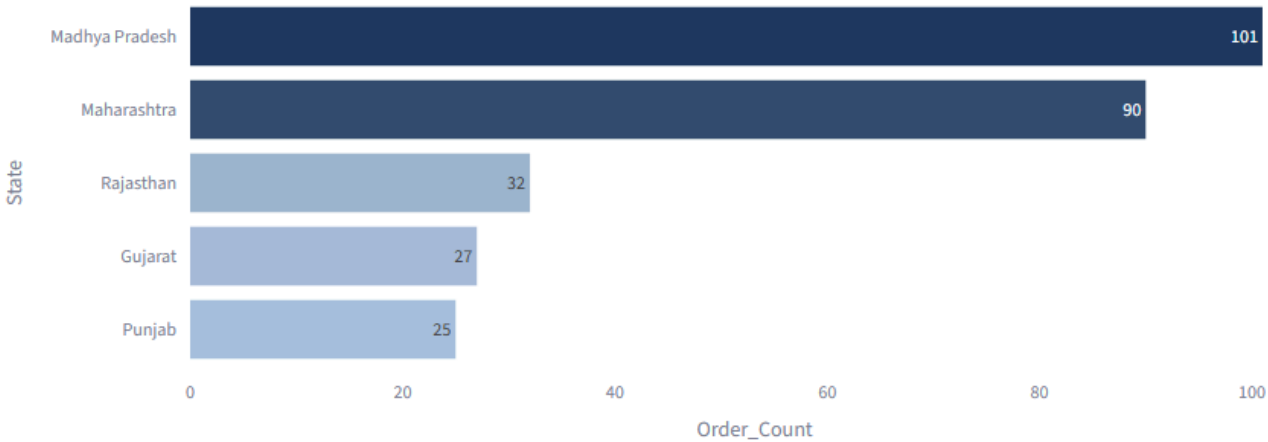


Figure 3 — Order volume across the top 5 states

Profit Margin — Top 5 States by Volume

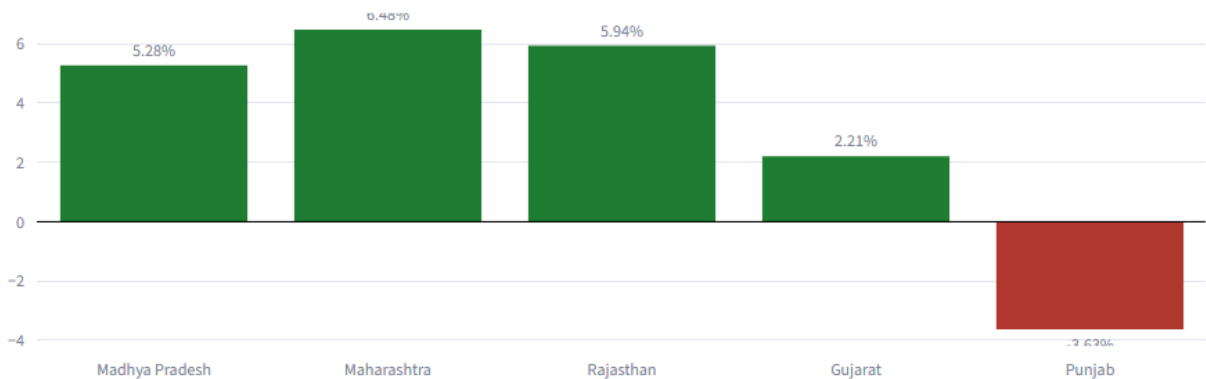


Figure 4 — Profit margin comparison across the top 5 states by order volume

Regional disparities observed

- **Madhya Pradesh and Maharashtra are the volume engines:** together they contribute 191 of 500 orders (38%) and roughly 62% of the sales generated by the top-5 group. Maharashtra is also the most profitable high-volume state (6.48% margin, ₹68.62 avg. profit/order), making it the strongest overall performer.

- **Punjab is a red flag:** despite being a top-5 state by volume (25 orders), it is the only one of the five running at a net loss (₹609, -3.63% margin) — meaning growth in order count there is currently destroying value, not creating it.
- **Order volume does not predict profitability:** looking beyond the top 5, Tamil Nadu has the worst margin in the entire dataset (-36.41%) despite only 8 orders, while smaller-volume states like West Bengal (17.75%), Haryana (14.95%), and Uttar Pradesh (14.48%) are the most profitable per rupee sold — well ahead of any top-5 state. This is a real disparity: the states driving the most orders are not the states generating the best returns.
- **City-level concentration mirrors the state pattern:** Indore (76 orders) and Mumbai (68 orders) alone account for nearly 30% of all orders, both within Madhya Pradesh and Maharashtra respectively — so the top-line volume story is really a two-city story.

Regions to prioritise for improvement

- **Punjab (immediate fix):** diagnose the loss — likely a heavy discount/return pattern or an unfavourable product mix (e.g., high share of Tables/Electronic Games) — before scaling order volume further there.
- **Tamil Nadu (contain before it grows):** at -36.41% margin it is currently the worst-performing state in the country; even though volume is low today, this needs a root-cause fix before any expansion investment goes into that market.
- **West Bengal, Uttar Pradesh, Haryana, Kerala (scale up):** these states already prove the business can be highly profitable (13–18% margins) at moderate volume — they are strong candidates for increased marketing/distribution investment to convert their profitability into higher order volume, the same way Maharashtra has.

Question 2 — App Exploration

Based on hands-on use of the Jar app together with a review of user feedback across the Play Store, App Store, and independent reviews, here is what stands out — both what works well and what could be improved.

Five things Jar gets right

1. Radically low entry point (₹10 auto-save)

Letting users start with as little as ₹10/day removes the single biggest psychological barrier to investing for first-time savers — the feeling that you need a large lump sum to "start investing." This is the core insight the entire product is built around, and it's genuinely well executed.

2. Round-off / spare-change automation

Rounding UPI spends to the nearest ₹10 and auto-investing the difference means saving requires zero ongoing decision-making after setup. For a first-time saver, removing the need to "remember to save" is more valuable than any interest rate — it converts a one-time intent into an automatic habit.

3. Flexible auto-pay cadence (daily/weekly/monthly)

Letting users choose their save frequency (and adjust the minimum accordingly) lets the product fit around irregular income patterns — relevant for gig workers, students, and small-business owners who don't have a fixed monthly salary cycle.

4. Bundled financial services on the same rails

Once trust and a UPI-linked habit are established, Jar cross-sells personal loans, health insurance, and jewellery (via VIBA/Nek) inside the same app. For a user who has never engaged with a financial product before, discovering these adjacent services inside an app they already trust is a much lower-friction path than a cold outreach from a separate lender or insurer.

5. Gamified engagement layer (Spin & Win, Jar Coins, referrals)

Small dopamine loops — spins after each deposit, Jar Coins redeemable for a rewards "mystery box," and referral incentives — keep a low-involvement product (passive gold accumulation) feeling active and rewarding day to day, which likely helps retention for a category that otherwise has very little to "check" on a daily basis.

Five areas where Jar could improve

1. Pricing transparency at the point of purchase

A recurring theme across user reviews is confusion about how much of the paid amount actually converts to gold after spread, GST, and making charges are deducted — with some users reporting a meaningful gap between what they paid and what showed up as gold value. Showing a clear "₹X will buy you Y grams at today's rate, including all charges" breakdown before every purchase would directly address this trust gap.

2. Withdrawal speed and clarity

Multiple reviews describe withdrawal delays and a lack of clear status updates during the redemption process, which is especially damaging for a product whose core promise is "your money, safely growing, available when you need it." A visible, real-time withdrawal status tracker (similar to an order-tracking UI) would reduce anxiety and support tickets simultaneously.

3. Customer support responsiveness

Several reviews mention slow or templated responses on account/withdrawal issues, sometimes going unresolved for days. For a savings app holding real money, support responsiveness is a trust-critical feature, not just a service-quality one — even a basic in-app live-status ticket system would likely reduce review-level frustration substantially.

4. Referral and rewards reliability

Users have reported referral rewards not crediting correctly or referred users not appearing in their referral list. Since referrals are one of Jar's primary organic growth channels, bugs here don't just frustrate individual users — they quietly undermine a core acquisition loop.

5. Clarity on regulatory status and jewellery purchase terms

Digital gold itself currently sits outside direct RBI/SEBI oversight, and some users have been surprised by no-cancellation/no-return terms on jewellery purchases made through the platform. Since Jar's biggest asset is user trust, being more proactive (rather than reactive, in review responses) about what protections do and don't apply — and making return/cancellation terms visible before checkout, not after — would strengthen rather than dilute that trust.

Question 3 — Product Exploration: New Business Opportunities

Jar's real asset isn't gold — it's the automated, trusted, daily-touch savings habit it has built with 4 crore+ users who transact in small, frequent amounts. Every new opportunity below is designed to extend that same automation-and-trust engine into an adjacent need, rather than asking users to adopt a new behaviour from scratch.

1. Goal-based micro-investing beyond gold

Extend the same round-off/auto-pay engine to let users save toward named goals — a phone, a wedding, an emergency fund — split across gold, a liquid mutual fund, or a recurring deposit, chosen automatically based on the goal's time horizon. This keeps the mechanism identical (automated micro-amounts) while diversifying the destination, deepening engagement without asking users to learn a new interaction model.

2. "Jar Score" — a credit-readiness layer for the credit-invisible

Millions of Jar users are first-time formal savers with no credit history. A consistent, automated saving pattern over months is itself a strong behavioural signal of repayment discipline. Jar could build a proprietary savings-based credit score, used first to underwrite its own personal loans more cheaply for consistent savers, and eventually licensed or shared (with consent) with partner lenders — turning saving behaviour into a tangible credit-access benefit.

3. Family & group savings pots

Many of Jar's core savers are supporting family goals (a child's education, a parent's medical fund, a sibling's wedding) rather than purely personal ones. A shared savings pot — where multiple contributors auto-pay into one visible goal — matches how Indian households actually pool money today (informally, via cash or chit funds) and gives Jar a natural entry into the group/community-savings space with very little new infrastructure required.

4. Bite-sized insurance and protection products

Jar has already validated appetite for this via its existing health insurance offering. This can be extended to low-premium, automated micro-insurance — accident cover, gold-specific theft/loss cover for physical redemptions, or even a small "gold-linked" life cover — sold in the same ₹10-₹50/day increments users are already comfortable auto-paying, rather than the lump-sum annual premiums that typically deter first-time insurance buyers.

5. Merchant and gig-worker savings-as-a-service

Small merchants and gig workers who already accept UPI have irregular income and almost no structured saving habit. A merchant-facing version of Jar's auto-save engine — automatically diverting a small percentage of each UPI payment received into savings — would extend the same core product to the supply side of UPI transactions, not just the consumer side, opening an entirely new user segment with the same underlying technology.

How Jar can execute this without diluting its core value proposition

- **Automation-first, not app-redesign-first:** every opportunity above reuses the existing auto-pay/round-off engine as the delivery mechanism, so users experience it as "more of what Jar already does for me" rather than a new product to learn.
- **Trust before monetisation:** given the pricing-transparency and support concerns identified in Question 2, Jar should close those trust gaps in its core gold product before layering new financial products on top — new offerings inherit the reputation of the existing ones.
- **Segment by income regularity, not just demographics:** salaried users, gig workers, and small merchants have different cash-flow shapes; the same automation engine should offer different default cadences/products per segment rather than a one-size-fits-all flow.
- **Milestone-based trust building:** frame growth as a ladder — first automated gold saving, then goal-based diversification, then credit-readiness, then protection products — so each milestone is unlocked by demonstrated saving behaviour, reinforcing the brand's core promise of gradual, safe financial progress rather than aggressive cross-selling.